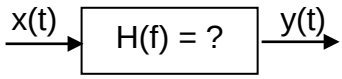
Non-parametric SYSTEM identification - Intro

What: Lecture #8: Spectral estimation of SIGNAL (non-parametric)
Lecture #9: Spectral estimation of SIGNAL (parametric)
- where signal x(t) is assumed to be measured without measurement noise

Lecture #10: Spectral estimation of SYSTEM (system identification)



- where x(t) and/or y(t) can only be measured including measurement noise
- non-parametric estimation of H(f)
- in this context often called FRF, Frequency Response Function
- non-parametric estimation often first step, e.g. recognize peaks etc.)

Lecture #11 + #12: Adaptive systems (including adaptive system identification)

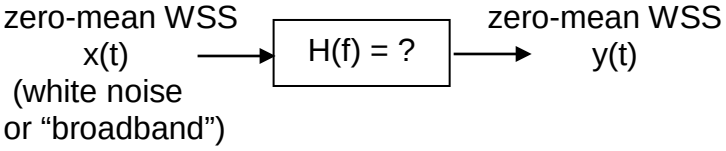
Why: Many applications in communication, acoustics, control, vibration etc., e.g.
- estimation of transmission channel's linear distortion
- estimation of electrical to acoustic transfer function (or vice versa)
- estimation of control system "plant"

How: Classical methods

Frequency sweep: $x(t) = \cos(\omega t), \quad \omega \in [\omega_a, \omega_b]$
$y(t) = |H(\omega)| \cos(\omega t + \angle H(\omega))$
sweep rate must be low enough to ensure all transients have died out at each frequency

Impulse response: $x(t) = "\delta(t)" \Rightarrow y(t) = h(t)$
very little energy in output

Set-up for <u>ideal</u> stochastic method



x(t) and y(t) can be measured <u>without</u> measurement noise

<u>Correlation functions and PSD's</u> (lecture #2-3 review):

$R_{xx}(\tau) = E[x(t)x(t+\tau)]$
$S_{xx}(f) = \mathcal{F}\{R_{xx}(\tau)\} = \lim_{T\to\infty} \frac{1}{T} E[|X_T(f)|^2]$
Likewise for y

$R_{xy}(\tau) = E[x(t)y(t+\tau)]$
$S_{xy}(f) = \mathcal{F}\{R_{xy}(\tau)\} = \lim_{T\to\infty} \frac{1}{T} E[X_T^*(f)Y_T(f)]$

<u>Basic I/O relations</u> (lecture #3 review):

$S_{xy}(f) = H(f)S_{xx}(f)$
$S_{yx}(f) = S_{xy}^*(f) = H^*(f)S_{xx}(f)$
$S_{yy}(f) = |H(f)|^2 S_{xx}(f) = H^*(f)S_{xy}(f) = H(f)S_{yx}(f)$

<u>Coherence function</u> (lecture #3 review):

$$\gamma_{xy}^2(f) \stackrel{\text{def}}{=} \frac{|S_{xy}(f)|^2}{S_{xx}(f)S_{yy}(f)} = \frac{S_{xy}(f) \big/ S_{xx}(f)}{S_{yy}(f) \big/ S_{xy}^*(f)} = \frac{H(f)}{H(f)} = 1$$

(compare from STAT: $r_{xy}^2 = \frac{\sigma_{xy}^2}{\sigma_x^2\sigma_y^2}$)

$0 \le \gamma_{xy}^2(f) \le 1$ measures to what degree (percentage) y(t) is generated by linear filtering of x(t) at frequency f

$\gamma_{xy}^2(f) < 1$ **if**
- x(t) and/or y(t) contains extraneous measurement noise
- H is non-linear
- more ("hidden") inputs than x(t)

Notation

In this context is often used one-sided PSD's
$G_{xx}(f) = 2\,S_{xx}(f)\,u(f) = \begin{cases} 2S_{xx}(f) & f \ge 0 \\ 0 & f < 0 \end{cases}$
likewise for $G_{yy}(f)$ and $G_{xy}(f)$